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RGB-Event based Pedestrian Attribute Recognition: A Benchmark Dataset and An Asymmetric RWKV Fusion Framework

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Abstract

Existing pedestrian attribute recognition methods are generally developed based on RGB frame cameras. However, these approaches are constrained by the limitations of RGB cameras, such as sensitivity to lighting conditions and motion blur, which hinder their performance. Furthermore, current attribute recognition primarily focuses on analyzing pedestrians' external appearance and clothing, lacking an exploration of emotional dimensions. In this paper, we revisit these issues and propose a novel multi-modal RGB-Event attribute recognition task by drawing inspiration from the advantages of event cameras in low-light, high-speed, and low-power consumption. Specifically, we introduce the first large-scale multi-modal pedestrian attribute recognition dataset, termed EventPAR, comprising 100K paired RGB-Event samples that cover 50 attributes related to both appearance and six human emotions, diverse scenes, and various seasons. By retraining and evaluating mainstream PAR models on this dataset, we establish a comprehensive benchmark and provide a solid foundation for future research in terms of data and algorithmic baselines. In addition, we propose a novel RWKV-based multi-modal pedestrian attribute recognition framework, featuring an RWKV visual encoder and an asymmetric RWKV fusion module. Extensive experiments are conducted on our proposed dataset as well as two simulated datasets (MARS-Attribute and DukeMTMC-VID-Attribute), achieving state-of-the-art results. The source code has been released on: https://github.com/Event-AHU/OpenPAR

1. Introduction

Existing Pedestrian Attribute Recognition (PAR) [38, 43–46, 54] systems are typically developed using RGB frame cameras, with the goal of enabling a model to accurately select matching attributes from a predefined list based on a given pedestrian image or video. Common attributes include *appearance information* such as hairstyle, gender, upper and lower body clothing, and carried objects. With the advancement of artificial intelligence technology, pedestrian attribute recognition has garnered increasing attention and is now widely applied in various human-centric visual tasks, including pedestrian detection and tracking [22, 23, 37, 42], person re-identification [9, 36, 52], and text-image retrieval [5, 8, 50, 59] for pedestrians. Despite significant progress, the overall performance of these systems remains constrained by the inherent limitations of RGB cameras, such as sensitivity to lighting conditions, clutter background, and susceptibility to motion blur. Interestingly, while multi-modal fusion has been extensively adopted in other computer vision tasks [29, 47, 49], it is surprising that pedestrian attribute recognition still predominantly relies on single-modality RGB cameras. This limitation highlights an opportunity for innovation through the integration of additional modalities to address the existing performance bottlenecks.

On the other hand, the pedestrian attributes defined in existing PAR benchmark datasets focus on describing the visible appearance cues, but none of them consider the invisible *emotion information*, which is also very important for human-centric perception. Recognizing emotions such as anxiety, anger, or joy can help the system better understand pedestrian behavioral intent and enhance the naturalness of human-computer interaction, improve safety and risk assessment, and support personalized services. Incorporating emotion analysis into pedestrian attribute recognition tasks can make the system smarter and more human-

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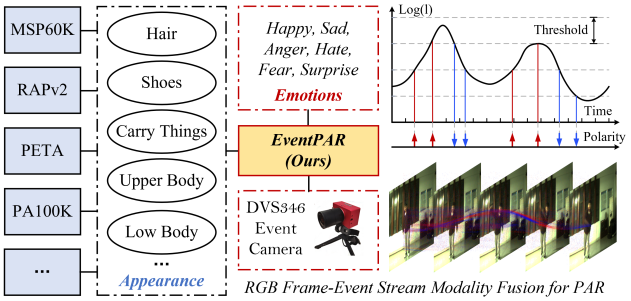


Figure 1. (left) Comparison between existing PAR datasets and our newly proposed EventPAR; (right) Imaging principle and results of Frame and Event cameras.

centric, while also providing support for a broader range of practical application scenarios.

In this work, we first propose to recognize human attributes using a DVS346 event camera that outputs aligned frame-event streams. This is because event cameras have inherent advantages in high dynamic range, high temporal resolution, and low power consumption. However, event streams contain some noise and are less sensitive to static objects. Therefore, combining them with RGB frame cameras is necessary to achieve more robust and accurate results. Due to the lack of datasets in this area, we first propose a large-scale multi-modal RGB-Event pedestrian attribute recognition benchmark dataset, termed **EventPAR**. This dataset covers common human appearance attributes as well as six human emotional attributes, totaling 50 attributes and 100K paired RGB-Event samples, as shown in Fig. 1. These videos were collected under various challenging scenarios, including different lighting conditions, motion blur, object occlusion, and adverse weather conditions. In addition to natural challenging factors, we also artificially introduced some adversarial attack noises and degradation strategies, such as various types of noise and occlusion. To provide a solid benchmark for future research, we re-train and report 17 mainstream PAR algorithms, as shown in Table 3.

Based on the proposed EventPAR dataset, we introduce a novel multi-modal RGB-Event pedestrian attribute recognition framework leveraging the RWKV model [6], which effectively integrates event streams to further enhance pedestrian attribute recognition results from the traditional RGB frame. Specifically, given the input RGB image and event streams, we first employ an RWKV encoder to extract visual features from both modalities. These features are then fed into an asymmetric RWKV fusion module, designed to better fuse spatial features from RGB frames with temporal information from event streams. This fusion module efficiently removes redundant event tokens while achieving interactive fusion with RGB features. Finally, a linear layer maps the fused features to multi-label attribute predictions. We extensively validate the proposed framework on three datasets and achieve state-of-the-art performance.

To sum up, we draw the main contributions of this paper as the following three aspects:

1). We propose to recognize the pedestrian attributes using an event camera and construct a large-scale benchmark dataset termed EventPAR. It contains 100K aligned RGB frames and event streams, collected from multiple scenarios, illuminations, etc.

2). We propose a novel RWKV-based framework for frame-event fusion based attribute recognition. A novel asymmetric RWKV fusion module fuses the RGB frame and event streams effectively and efficiently.

3). We build a large-scale benchmark for the EventPAR dataset by re-training 17 representative and strong PAR algorithms. This benchmark will be a good platform to boost the development of event-based attribute recognition further.

In addition, we also conduct extensive experiments on three PAR benchmark datasets, including the newly proposed EventPAR, MARS-Attribute, and DukeMTMC-VID-Attribute datasets. More experiments fully validated the effectiveness of our proposed model for the frame-event pedestrian attribute recognition.

2. Related Works

2.1. Pedestrian Attribute Recognition

Pedestrian Attribute Recognition aims to identify and classify multiple attributes of pedestrians, such as age, gender, clothing type, and accessories, from images or videos. Video-based PAR is a relatively new research topic. Researchers have introduced multi-task models, temporal pooling, and sparse temporal attention modules, among other techniques. Chen et al. [2] proposed a multi-task model with an attention module to handle each attribute per frame. Specker et al. [32] introduced global features to incorporate information from different frames. Li et al. [16] proposed a sparse temporal attention module to select unobstructed frames, enhancing robustness. Thakare et al. [34] calculated the mutual correlations of attribute predictions from pedestrian images across different views, while Liu et al. [25] introduced a spatio-temporal saliency module to capture correlations between attributes in both spatial and temporal domains. Different from these works, this paper firstly exploits the multi-modal PAR task using the event camera and proposes a large-scale benchmark dataset and RWKV-based framework for this task.

2.2. Event-based Vision

Event cameras have been widely used in computer vision and multi-modal related tasks. Specifically, in the image

restoration and enhancement tasks, EventSR [35] proposes a novel end-to-end super-resolution image reconstruction framework that reconstructs low-resolution images from event streams, enhances image quality, and upsamples the enhanced images. EG-VSR [26] learns spatial-temporal coordinates and implicit neural representations from RGB frames and event features, leveraging the high temporal resolution of events to achieve video super-resolution at arbitrary resolutions. Jiang et al. [13] and colleagues have introduced a method that effectively restores motion-blurred video sequences from event-based camera data, achieving this by integrating global and local scale visual and temporal knowledge, along with a differentiable directional event filtering module. Jiang et al. [12] proposed a posterior attention module that adjusts standard attention using the prior knowledge provided by event data, which improves the segmentation performance for moving objects. Inspired by these works, in this paper, we attempt to achieve high-performance PAR by fusing the event stream and RGB frames.

2.3. RWKV Model

The Receptance Weighted Key Value (RWKV) [27] was originally proposed in natural language processing to address the quadratic time complexity of Transformers in handling long sequences. Inspired by the linear attention mechanism in the Attention Free Transformer (AFT) [31], RWKV introduced the WKV attention mechanism. At the same time, the token shift operation was proposed to capture local context. This combination allows RWKV to take advantage of both the parallel processing attention mechanism of Transformers and the sequential time-series pattern of Recurrent Neural Networks (RNNs). Later, RWKV was extended to computer vision tasks. BSBP-RWKV [48] proposes an efficient medical image segmentation framework that integrates Perona-Malik Diffusion for noise suppression with boundary preservation and an RWKV-based architecture to reduce computational complexity. RWKV-SAM [53] proposes an efficient segment-anything framework with a mixed convolution-RWKV backbone and multiscale token decoder, using a joint training benchmark to achieve twice the inference speed and superior segmentation quality compared to Transformer and vision Mamba baselines. Unlike these works, this paper is the first to apply RWKV to pedestrian attribute recognition.

3. EventPAR Benchmark Dataset

3.1. Protocols

In this work, we follow the guidelines below when constructing the EventPAR benchmark dataset: ***1). Multi-Modal:*** As shown in Table 1, unlike existing frame-based PAR datasets, we utilize the DVS346 camera for data collection, obtaining spatiotemporal aligned multi-modal data comprising visible light and event streams. It is the first multi-modal PAR database, laying a solid data foundation for subsequent research. ***2). Emotion Attributes:*** Unlike conventional PAR datasets that primarily focus on pedestrians' appearance attributes, our new dataset incorporates six fundamental human emotional attributes, i.e., *Happiness, Sadness, Anger, Surprise, Fear,* and *Disgust*. This addition effectively fills the gap in the attribute recognition domain by addressing the lack of discussion on the emotional analysis dimension. The static and dynamic presentations are shown on the right side of Fig. 2. ***3). Diverse Scenes and Cross-Seasonal:*** Different from existing PAR datasets that involve short-term data collection, our data acquisition spanned several months, encompassing data from different seasons (summer and winter). Additionally, it covers a variety of scenes and weather conditions, such as daytime and nighttime scenes, as well as weather conditions like sunny and rainy days, as shown in Fig. 2 (a, b). These elements significantly enhance the diversity and challenge of the dataset, providing better validation of a model's fitting ability and generalization capability. ***4). Large Scale:*** It contains 100K spatio-temporally aligned pedestrian samples of RGB frames and event streams, 12 attribute groups, and 50 fine-grained pedestrian attributes. It is one of the largest PAR datasets, comparable to PA100K in scale, as shown in Table 2. ***5). Simulated Complex Real-world Environments:*** Our dataset is carefully designed to incorporate a wide range of challenges, such as variations in illumi-

Table 1. Comparison with existing PAR benchmark datasets. #**Att.** is short for the number of Attributes. Deg and Adv is short for Degradation and Adversary, respectively.

Dataset	Year	#Att.	#Samples	Video	Frame	Event	Emotion	Deg	Adv
PETA [4]	2014	61	19,000	✗	✓	✗	✗	✗	✗
WIDER [21]	2016	14	57,524	✗	✓	✗	✗	✗	✗
RAPv1 [18]	2016	69	41,585	✗	✓	✗	✗	✗	✗
PA100K [24]	2017	26	100,000	✗	✓	✗	✗	✗	✗
RAPv2 [19]	2019	76	84,928	✗	✓	✗	✗	✗	✗
MSP60K [15]	2024	57	60,015	✗	✓	✗	✗	✓	✗
MARS-Attribute [56]	2016	43	16,360	✓	✗	✗	✗	✗	✗
DukeMTMC-VID [28]	2017	36	4,832	✓	✗	✗	✗	✗	✗
EventPAR (Ours)	2025	50	100,000	✓	✓	✓	✓	✓	✓

Table 2. Attribute groups and details defined in our proposed EventPAR dataset.

Attribute Group	Details
#01 Gender	Male, Female
#02 Age	Child, Adult, Elderly
#03 Body Size	Fat, Normal, Thin
#04 Viewpoint	Front, Back, Side
#05 Head	Long Hair, Black Hair, Hat, Glasses, Mask, Scarf, Headphones
#06 Upper Body	Short Sleeves, Long Sleeves, Shirt, Coat, Cotton-padded Coat
#07 Lower Body	Trousers, Shorts, Jeans, Long Skirt, Short Skirt, Casual Pants, Dress
#08 Shoes	Casual Shoes, Other Shoes
#09 Accessory	Backpack, Shoulder Bag, Hand Bag, Umbrella, Cup, Cellphone
#10 Posture	Walking, Running, Standing, Sitting
#11 Activity	Smoking, Reading
#12 Emotion	Happiness, Sadness, Anger, Surprise, Fear, Disgust

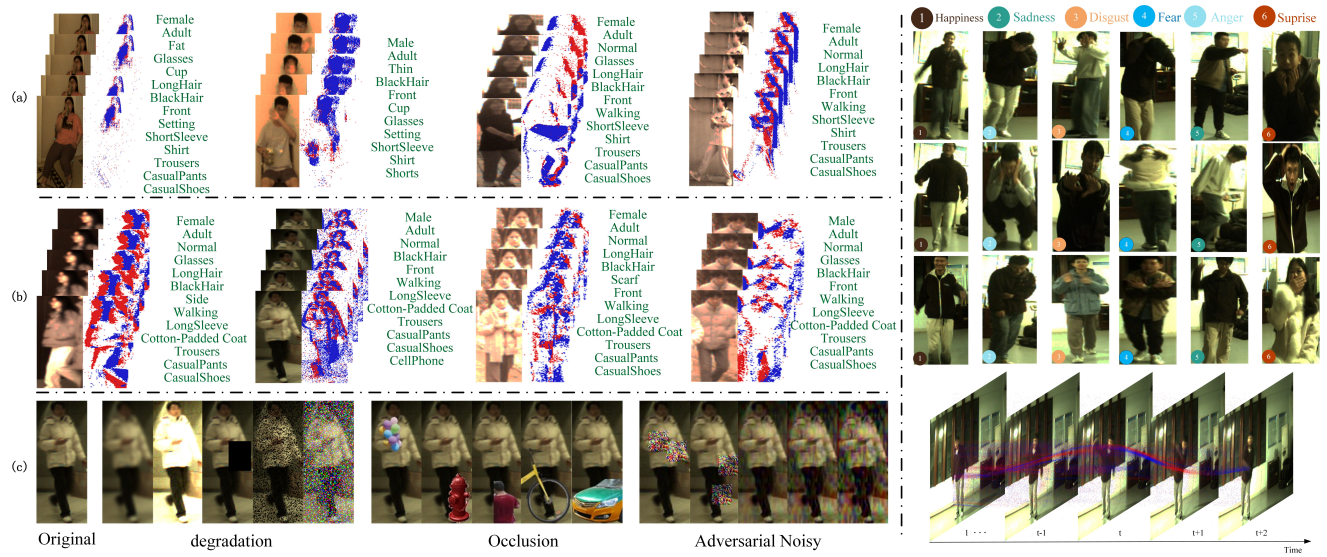


Figure 2. An illustration of representative samples in our newly proposed EventPAR dataset. On the left, subfigures (a) and (b) illustrate RGB and Event bimodal data collected under summer and winter conditions, across different genders and indoor/outdoor environments. Subfigure (c) shows examples with various types of added noise. On the right, we visualize the six proposed emotion attributes under both static and dynamic scenarios.

nation, motion blur, object occlusion, and adverse weather conditions, all of which simulate the complex real-world difficulties encountered in pedestrian attribute recognition. To further increase the complexity and practical relevance of the dataset, we employ *adversarial attack* techniques [1] during data processing, adding an extra layer of difficulty, as shown in Fig. 2 (c).

Moreover, EventPAR exhibits a real-world long-tail distribution, as shown in Fig. 3 (a), where frequent attributes include *Adult, Back Hair,* and *Normal Size* while rare ones such as *Smoking, Child,* and *Elderly* appear less often. The attribute co-occurrence matrix is shown in Fig. 3 (b).

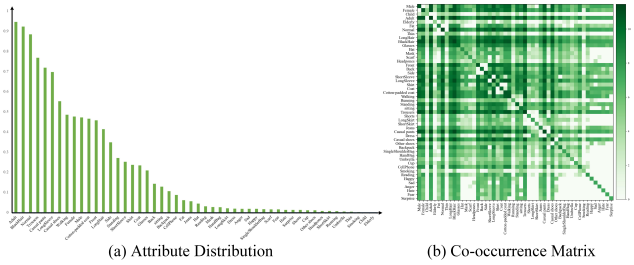


Figure 3. (a) Attributes Distribution: Bar graph showing the prevalence of individual attributes across the dataset; (b) Co-occurrence Matrix of Attributes: Logarithmic heatmap showing the co-occurrence frequency of attribute pairs.

3.2. Benchmark Baselines

Based on our newly proposed dataset, we retrained and tested 17 mainstream PAR models, providing a solid baseline for future work on the EventPAR dataset. Specifically, the following algorithms were included: *1). CNN-based:* DeepMAR [17],Zhou et al. [57], RethinkPAR [11], SSC-Net [10], SSPNet [30]. *2). Transformer-based:* DFDT [55], PARFormer [7]. *3). Mamba-based:* MambaPAR [41], MaHDFT [40]. *4). Human-Centric Pre-Training Models for PAR:* PLIP [60]. *5). Visual-Language Models for PAR:* VTB [3], Label2Label [20], PromptPAR [39], SequencePAR [14].

4. Methodology

4.1. Overview

To improve recognition accuracy and robustness, this paper proposes a video-based multimodal pedestrian attribute recognition framework shown in Fig. 4. Event data is used to supplement RGB information loss, enhancing stability. Features are extracted via Visible-RWKV and Event-RWKV encoders, fused through the OTN-RWKV module, and then classified for attribute prediction. Please refer to the following for more detailed information.

4.2. Network Architecture

• **Input Encoding:** In this work, event modality is used to assist the RGB video in overcoming information loss

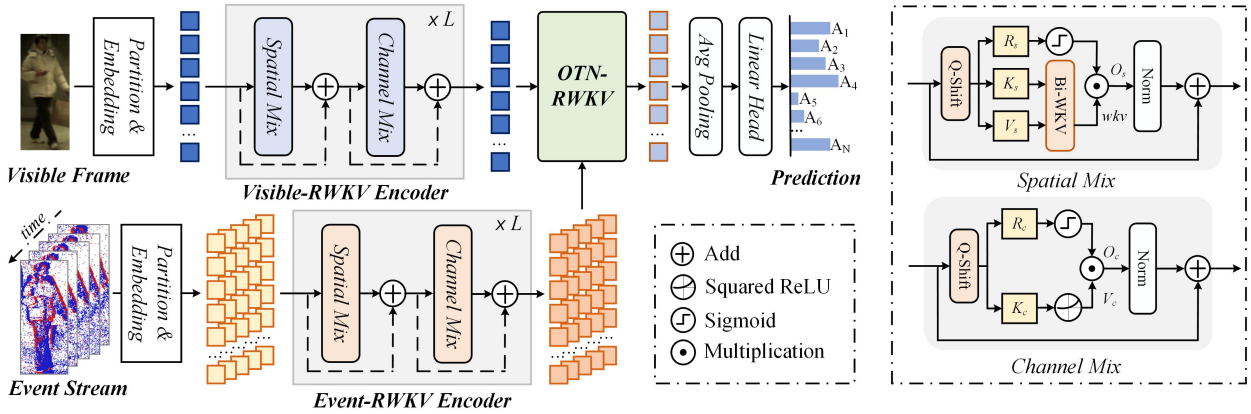


Figure 4. An overview of our proposed OTN-RWKV guided RGB-Event fusion for pedestrian attribute recognition.

in complex environments. Given a sequence of an RGB video denote as $X_r \in \mathbb{R}^{T\times H\times W\times C}$, where T is the total number of frames, and H, W, C correspond to the height, width, and the number of channels of each frame, respectively. We take event streams as $\mathcal{E} = \{e_1, e_2, ..., e_M\}$, each event point e_j can be denoted as $\{x, y, t, p\}$, representing spatial coordinates, timestamps, and polarity, respectively. To align with the RGB sequence, we stack the event streams into event frames that are temporally synchronized with the RGB frames based on exposure times. Therefore, we can obtain the event frame sequence $X_e \in \mathbb{R}^{T\times H\times W\times C}$.

By inputting a sequence of RGB frames $X_r \in \mathbb{R}^{T\times H\times W\times C}$ and a corresponding sequence of event frames $X_e \in \mathbb{R}^{T\times H\times W\times C}$, we first apply a partitioning operation to process data for both modalities, transforming each frame into HW/p^2 patches, where p denotes the patch size. Then, the patches are added with Position Embedding (P.E.), which are used to encodes spatial information to obtain RGB tokens $X_r \in \mathbb{R}^{T\times N\times C}$ and event tokens $X_e \in \mathbb{R}^{T\times N\times C}$ separately, where N is the number of tokens.

• **RWKV Encoder:** In this paper, we employ Vision-RWKV [6] as our visual encoder, which adopts a block-stacked design with L layers to enhance its encoding capability, where each block consists of a Spatial-Mix module and a Channel-Mix module. In each layer, RGB tokens $X_r \in \mathbb{R}^{T\times N\times C}$ and event visual tokens $X_e \in \mathbb{R}^{T\times N\times C}$ are first fed into the Spatial-Mix module, which performs linear complexity global attention computation. Specifically, as shown in the top right of Fig. 4, the input visual tokens first undergo a token Shift operation to capture local context information before being fed into three parallel linear layers to obtain the matrices R_s,K_s,$V_s \in \mathbb{R}^{T*N\times C}$. The formulas are as follows:

$$\begin{aligned} R_\mathrm{s} &= \text{Q-Shift}_R(X_i)W_R, \\ K_\mathrm{s} &= \text{Q-Shift}_K(X_i)W_K, \\ V_\mathrm{s} &= \text{Q-Shift}_V(X_i)W_V, \quad i \in (r, e) \end{aligned} \tag{1}$$

where the Q-Shift function is a specialized token shift mechanism specifically designed for vision tasks. After that, K_s and V_s are used to calculate the global attention result $wkv \in \mathbb{R}^{T*N\times C}$ by the Bi-WKV module, which is a linear complexity bidirectional attention mechanism. Subsequently, the wkv is multiplied with $\sigma(R)$, which controls the output O_i probability,

$$\begin{aligned} O_i &= (\sigma(R_s) \odot wkv)W_O, \\ wkv &= \text{Bi-WKV}(K_s, V_s), \quad i \in (r, e), \end{aligned} \tag{2}$$

where the operator σ represents the sigmoid function, while $\odot$ denotes element-wise multiplication. After the output linear projection, the features are subjected to layer normalization to enhance stability and facilitate effective training.

Subsequently, the tokens are fed into the Channel-Mix module to achieve channel-wise fusion, as shown in the bottom right of Fig. 4. The parameters R_c and K_c are computed similarly to the Spatial-Mix before the final output,

$$\begin{aligned} R_c &= \text{Q-Shift}_R(O_i)W_R, \\ K_c &= \text{Q-Shift}_K(O_i)W_K, \end{aligned} \tag{3}$$

the gating mechanism $\sigma(R_c)$ is used to modulate the output O_i'. To maintain information integrity and promote gradient flow, residual connections are introduced to fuse the module outputs, thereby effectively alleviating the vanishing gradient problem. This process can be expressed as:

$$\begin{aligned} O_i' &= (\sigma(R_c) \odot V_c)W_O, \\ V_c &= \text{SquaredReLU}(K_c)W_V, \end{aligned} \tag{4}$$

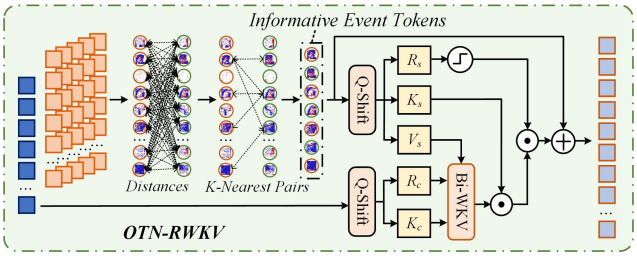


Figure 5. Illustration of our proposed OTN-RWKV for RGB-Event fusion.

• **OTN-RWKV:** Due to the asynchronous nature of event cameras, event frames contain a large amount of redundant information, increasing computational burden and reducing the effectiveness of auxiliary data. To address this, we propose the OTN-RWKV fusion module, which uses a similarity matrix to identify the top-K most similar token pairs from the event tokens output O_e', retaining only the most representative tokens. This method improves computational efficiency and enhances the quality of multimodal fusion. The implementation process is as follows:

$$O_e^{''} = \text{KNP}_{filter}(\text{sim}(O_e', O_e')) \odot O_e', \tag{5}$$

where $O_e^{''} \in \mathbb{R}^{N' \times C}$ represents the tokens for the informative filtered event. After filtering, the number of tokens in both modalities is aligned to facilitate fusion.

Although traditional fusion methods (e.g., concatenation, addition, and 1×1 convolution) offer strong compatibility, they often cause information loss, limiting the preservation of fine-grained multimodal details. Inspired by the concept of cross attention and LCR[51], in this work, we further propose a novel interactive fusion strategy. As shown in Fig. 5, we first apply the Q-shift operation separately to the two types of visual tokens, corresponding to Eq.1 and Eq. 3, respectively. Subsequently, V_s, R_c, and K_c are used to compute the cross-attention results through a linear complexity bidirectional attention mechanism,

$$O_{fusion} = \sigma(R_s) \odot LN(K_s \odot \text{Bi-}WKV) + O_e^{'}, \tag{6}$$

$$\text{Bi-}WKV(R,K,V)_t = \frac{\sum_{i=0,i\neq t}^{M} e^{-(|t-i|-1)/M\cdot v_s + r_i} k_i + e^{u+r_t} k_t}{\sum_{i=0,i\neq t}^{M} e^{-(|t-i|-1)/M\cdot v_s + r_i} + e^{u+r_t}}. \tag{7}$$

where t represents the t-th token of the input, M represents the total number of tokens, u is a C-dimensional learnable vector that acts as a bonus indicating the current token, and LN refers to LayerNorm. Unlike LCR[51], which uses the learnable vector w, we replace w with the output V_s from Eq. 1 to better focus on the current sample.

4.3. Loss Function

After the effective fusion of the two modalities using the OTN-RWKV module, the resulting representation is fed into an average pooling layer and then a linear classification head, yielding the final attribute prediction P_{attr}. In the training phase, we adopt the weighted cross-entropy loss (WCE Loss)[3] to alleviate the distribution imbalance between pedestrian attributes, which is widely used by attribute recognition,

$$\mathcal{L}_{wce} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{M} \omega_j \left(y_{ij}\log(p_{ij}) + (1-y_{ij})\log(1-p_{ij})\right), \tag{8}$$

here, N represents the total number of samples, M denotes the number of attributes, and ω_j is the weight for the j-th attribute. y_{ij} and p_{ij} denote the ground truth and predicted result, respectively.

5. Experiments

5.1. Datasets & Evaluation Metric

This study benchmarks 17 pedestrian attribute recognition methods and compares our method with state-of-the-art approaches on the benchmark and public datasets MARS-Attribute [56] and DukeMTMC-VID-Attribute [28]. For the evaluation of our and the compared PAR models, we use five widely adopted evaluation metrics to assess performance, including: **mean Accuracy** (mA), **Accuracy** (Acc), **Precision** (Prec), **Recall**, and **F1-score** (F1).

5.2. Comparison on Public Datasets

In this section, we conduct a detailed comparison between our model and existing pedestrian attribute recognition (PAR) algorithms on three benchmark datasets. Since the MARS-Attribute and DukeMTMC-VID-Attribute datasets contain only the RGB modality, we simulate the corresponding event data to ensure the completeness and fairness of the experiments.

• **Results on EventPAR Datasets.** We collect and analyze public PAR methods from 2015 to 2025 on the EventPAR dataset as shown in Table 3, methods such as SequencePAR [14], and PARFormer [7] have demonstrated superior performance. Our method achieves good results using only RGB data, with mA, Acc, and F1 scores of 79.32, 76.00, and 83.22 respectively. Adding the event modality further improves these to 87.66, 84.78, and 89.07, achieving the best overall performance and demonstrating the significant benefit of event data for model robustness in complex environments.

• **Results on MARS-Attribute Datasets [56].** As shown in Table 4, comparing our method with several state-of-the-art approaches. Our method achieves the best performance on most evaluation metrics, achieving 73.21, 85.63, 81.53, and 83.22 in Accuracy, Precision, Recall, and F1, respectively. These results reflect a balanced performance with high accuracy and effective attribute capture.

Table 3. Comparison with public methods on our datasets. Size indicates the size of checkpoint file, and #**MS** means the GPU memory used for optimization, Test time refers to inference time.

Methods	**Publish**	**mA**	**Acc**	**Prec**	**Recall**	**F1**	**Test Time**	**Params** (MB)	**Flops** (GB)	**Size** (MB)	#**MS** (MB)
#01 DeepMAR [17]	ACPR15	66.57	69.53	74.90	88.54	81.57	64s	23	25	181	592
#02 ALM [33]	ICCV19	57.18	64.17	75.59	73.20	74.38	813s	17	8	66	785
#03 Strong Baseline [11]	arXiv20	73.75	61.86	67.23	80.78	75.43	47s	24	25	91	576
#04 RethinkingPAR [11]	arXiv20	81.37	80.84	86.31	87.57	86.93	44s	24	24	91	580
#05 SSCNet [10]	ICCV21	63.10	66.07	72.72	83.22	77.62	49s	24	9	90	588
#06 VTB [3]	TCSVT22	88.41	83.83	87.89	89.31	88.53	243s	93	66	333	525
#07 Label2Label [20]	ECCV22	72.49	74.01	86.60	79.02	82.19	398s	66	37	829	650
#08 DFDT [55]	EAAI22	61.71	63.14	79.17	70.63	74.66	287s	88	37	669	2584
#09 Zhou et al. [57]	IJCAI23	56.46	60.89	73.37	73.62	73.50	42s	234	24	536	701
#10 PARFormer [7]	TCSVT23	83.12	80.48	85.14	88.41	86.53	438s	195	205	756	2481
#11 VTB-PLIP [60]	arXiv23	67.25	68.37	77.75	79.72	78.37	229s	31	22	591	501
#12 Rethink-PLIP [60]	arXiv23	68.75	70.03	81.82	78.04	79.89	37s	21	23	144	481
#13 PromptPAR [39]	TCSVT24	86.51	82.27	86.35	89.36	87.64	1312s	8	5	1200	2296
#14 SSPNet [58]	PR24	66.92	67.49	78.73	76.90	77.80	69s	40	26	100	1898
#15 MambaPAR [40]	arXiv24	50.01	42.32	54.81	57.31	55.63	105s	0.64	0.41	98	802
#16 MaHDFT [40]	arXiv24	50.43	44.98	59.10	59.70	58.57	292s	164	6033	499	1313
#17 SequencePAR [14]	PR25	86.27	84.42	88.81	89.12	88.83	1497s	466	577	1776	9901
OTN-RWKV (Ours, RGB Only)	-	79.32	76.00	82.37	84.55	83.22	68s	107	39	429	2200
OTN-RWKV (Ours, RGB+Event)	-	**87.70**	**84.94**	**89.15**	**89.48**	**89.18**	361s	201	77	836	2312

Table 4. Results on MARS-Attribute and DukeMTMC-VID-Attribute RGB-Event based PAR dataset.

Methods	**Backbone**	**MARS-Attribute Dataset**				**DukeMTMC-VID-Attribute Dataset**			
		Acc	**Prec**	**Recall**	**F1**	**Acc**	**Pren**	**Recall**	**F1**
VTB [3]	ViT-B/16	72.73	82.79	83.52	82.89	72.50	83.60	82.24	82.38
Zhou et al. [57]	ConvNext	69.43	81.75	80.64	81.19	65.51	78.42	77.40	77.91
VTB-PLIP [60]	ResNet50	54.93	70.02	69.23	69.14	43.95	60.80	58.52	59.10
Rethink-PLIP [60]	ResNet50	47.85	62.15	64.80	63.45	40.25	55.41	56.13	55.77
SSPNet [58]	ResNet50	65.68	77.09	79.74	78.39	67.53	78.54	79.78	79.15
MambaPAR [40]	Vim	69.07	81.83	79.39	80.24	61.55	75.54	74.84	74.24
OTN-RWKV(Ours)	RWKV-B	**73.21**	**85.63**	81.53	**83.22**	**73.15**	**84.45**	82.16	**82.78**

• **Results on DukeMTMC-VID-Attribute Datasets [28].** As shown in Table 4, we also compared our method with several mainstream approaches on this dataset, achieving strong results with accuracy, precision, recall, and F1 scores of 73.15, 84.45, 82.16, and 82.79, respectively. Our method shows stable performance across diverse data environments, confirming its effectiveness and adaptability, with strong application potential.

5.3. Ablation Study

In this section, we conduct a comprehensive ablation study on the core modules of EventPAR, including different input configurations, various backbone architectures, fusion strategies and event data aggregation methods. Other experiments are included in the supplementary materials.

• **Analysis of different input settings.** In this study, We varied RGB input counts and ran extensive experiments as shown in Table 5. RGB or Event alone performed well, with RGB slightly worse due to noise. Combining both improved results, showing complementarity. Too many RGB inputs hurt performance due to redundancy and noise, burdening feature extraction and weakening semantic focus.

Table 5. Comparison of different input settings.

RGB	**Event**	**mA**	**Acc**	**Prec**	**Recall**	**F1**
1	0	79.41	76.22	82.43	84.65	83.27
0	5	87.14	84.52	89.07	89.09	88.91
1	5	**87.70**	**84.94**	**89.15**	**89.48**	**89.18**
3	5	85.97	81.91	86.84	86.98	86.61
5	5	85.69	82.24	87.19	86.88	86.75

Table 6. Comparison of different feature fusion methods.

Method	**mA**	**Acc**	**Prec**	**Recall**	**F1**
Concat	87.63	84.60	88.80	89.40	88.96
Add	87.17	84.62	88.96	89.30	88.99
1×1 Conv	83.77	80.33	84.78	88.63	86.44
OTN-RWKV	**87.70**	**84.94**	**89.15**	**89.48**	**89.18**

Table 7. Comparison of different aggregation strategies.

No.	Max	Mean	GNN	Sim	mA	Acc	F1
01	✓				87.56	84.62	88.98
02		✓			87.50	84.68	89.01
03			✓		84.53	82.24	87.50
04				✓	**87.59**	**84.80**	**89.10**

• **Analysis of different feature fusion methods.** As shown in Table 6, We compared our proposed OTN-RWKV fusion strategy with three common methods: Concat, Add, and 1×1Conv. OTN-RWKV achieved top results with mA 87.70, ACC 84.94, and F1 89.18, outperforming other methods. It effectively fuses RGB and event features, boosting robustness and performance.

• **Analysis of different aggregation strategies of Event.** As shown in Table 7, we systematically compared four event data aggregation strategies: Max Pooling, Mean Pooling, a graph-based GNN, and our proposed Similarity-based approach. Our method adaptively fuses event features, preserves temporal cues, suppresses noise, and validates the effectiveness of the aggregation strategy.

• **Analysis of different backbone.** To validate the effectiveness of the RWKV, we conducted comparisons with ViT and ResNet-50 under the same settings. RWKV achieved superior performance with mA, Accuracy, and F1 of 87.70, 84.94, and 89.18, respectively, demonstrating its advantage in pedestrian attribute recognition, as shown in Table 8.

Table 8. Comparison of different backbones.

No.	Backbone	mA	Acc	F1
01	ViT	84.05	82.41	87.60
02	ResNet50	83.85	82.45	87.64
03	RWKV	**87.70**	**84.94**	**89.18**

5.4. Visualization

• **Visualization of Predicted Results.** As illustrated in Fig. 6, We visualize pedestrian attribute prediction under different modalities. Single-modal inputs struggle in challenging conditions, while bimodal fusion effectively exploits complementary cues, enhancing recognition accuracy and robustness.

• **Visualization of Emotion Attribute Prediction.** Fig. 7 presents a comparison of six emotion attributes across multiple metrics. The proposed RGB-Event fusion effectively captures fine-grained emotional cues and consistently outperforms single-modal baselines, confirming the importance of emotional information in enhancing pedestrian recognition.

We visualize the event aggregation strategy, details are provided in the supplementary materials.

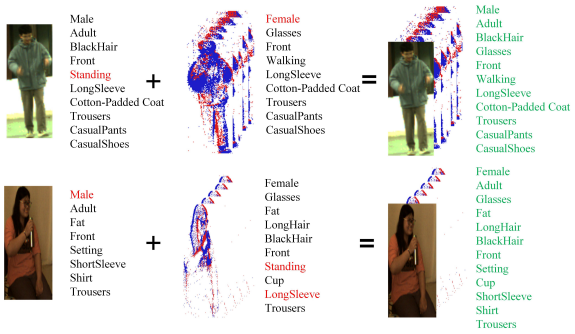


Figure 6. Visualization of pedestrian attribute prediction results.

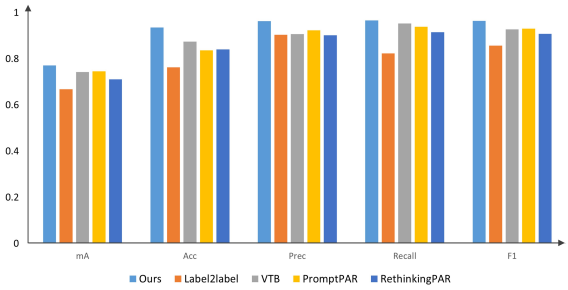


Figure 7. Visualization of emotion attribute prediction results.

6. Conclusion and Future Works

In this paper, we addressed the limitations of existing pedestrian attribute recognition (PAR) methods, which are predominantly based on RGB frame cameras and suffer from challenges such as sensitivity to lighting conditions and motion blur. Moreover, current approaches focus primarily on external appearance and clothing attributes, neglecting the emotional dimensions of pedestrians. To overcome these constraints, we introduced a novel multi-modal RGB-Event attribute recognition task, leveraging the advantages of event cameras in low-light, high-speed, and low-power scenarios. Our key contribution is the creation of EventPAR, the first large-scale multi-modal PAR dataset, which includes 100K paired RGB-Event samples covering 50 attributes related to both appearance and six human emotions. By retraining and evaluating mainstream PAR models on EventPAR, we established a comprehensive benchmark, offering valuable data and algorithmic baselines for the community. Furthermore, we proposed a novel RWKV-based multi-modal PAR framework, incorporating an RWKV visual encoder and an asymmetric RWKV fusion module. Extensive experiments conducted on EventPAR and two simulated datasets demonstrated state-of-the-art performance, underscoring the effectiveness of our approach. In our future works, we will focus on exploiting learning-based new event representations for the perception of event streams for pedestrian attribute recognition.

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